

Srinivasan Poonkundran

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PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions

Sept. 2021 – Jan. 2024

Full Stack Engineer

Remote

- Built scalable Java Spring Boot microservices with AWS deployment and robust database management, and developed interactive Angular dashboards to visualize KPIs for business stakeholders.
- Spearheaded performance improvements by optimizing SQL queries and caching, reducing load times by 25%.
- Created a Python-based PDF label validator, saving \$1600 in annual manual effort, and collaborated with Agile teams to use data insights in product development and decision-making.
- Designed and launched a key product feature that improved customer satisfaction by 20%, and revamped UI/UX to boost engagement by 15%.
- Skills:** Python, AWS, SQL/NoSQL, Data Analytics & Visualization, Java, Spring Boot, Unit Testing, Angular, Microservices, CI/CD, Performance Optimization, Problem Solving, Docker, Agile, Cross-functional Collaboration

Clothing Store Billing and Management System – Local Retail Client

Jan. 2023 – Jan. 2024

Software Developer

Remote

- Built an Android app to manage invoices, sales, and staff for a clothing store with 1,000+ monthly transactions.
- Designed sales dashboards with charts to track daily revenue, top-selling items, and seasonal trends.
- Implemented barcode-based billing to automate checkout processes which reduced average billing time from 3 minutes to 90 seconds per transaction and cut manual entry errors from 20 per week to fewer than 1.
- Used AWS S3 to store invoices and records securely, enabling cloud backup and cross-device access.
- Skills:** Android, Java, AWS S3, SQLite, Data Visualization, WhatsApp API, UI/UX Design, RegEX

PROJECTS

GradeLens – RAG Powered Auto Grader | [GitHub](#)

- Developed an AI-driven grading platform leveraging Retrieval Augmented Generation (RAG) to automate and contextualize student paper evaluation.
- Designed scalable Django-based backend services and retrieval pipelines using pgvector for document ingestion, rubric application, and grading inference from professor-uploaded course materials (PDF/PPT).
- Built a React-based professor dashboard for rubric customization and review, achieving 90% parity with human evaluators while significantly reducing grading time.
- Skills:** Retrieval Augmented Generation (RAG), NLP, Hugging Face, Python, Django REST framework, React

Emotion Detection from Text | [GitHub](#)

- Developed a Natural Language Processing (NLP) system to classify English text into seven emotions (anger, joy, sadness, fear, surprise, disgust, neutral), extending sentiment analysis beyond basic polarity detection.
- Implemented a BERT-based embedding model (bert-base-uncased) with frozen transformer layers and a Bidirectional GRU classifier in Keras, achieving an F1-micro score of 0.83 on the validation set.
- Optimized model performance using TensorFlow's mixed precision, EarlyStopping, and ModelCheckpoint, reducing training time from 6 minutes to 3-4 minutes and enabling efficient real-time inference.
- Skills:** BERT, Hugging Face, TensorFlow, Keras, GRU, Natural Language Processing (NLP), Python

Cricket Metrics | [GitHub](#) | [Website](#)

- Built a Shiny app cricket analytics dashboard to visualize performance from 150,000+ data points.
- Simulated live match data using APIs and integrated the output with a Quarto-based interactive frontend.
- Designed and implemented data visualizations, including line charts, bar charts, donut charts, and map plots, to provide insights into player performance, scoring trends, partnerships, and bowling economy rates.
- Skills:** R, Shiny, Quarto, Java, Spring Boot, MySQL, REST API, Data Visualization, ggplot2, Dashboard Design

EDUCATION

The University of Arizona

Tucson, AZ

Masters in Science, Data Science — Distinguished Graduate Scholar

Jan. 2024 – Dec. 2025

Savitribai Phule Pune University

Pune, India

Bachelor of Engineering, Computer

Aug. 2017 – May 2021

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL/NoSQL, MongoDB, JavaScript, Node.js, HTML/CSS, R/R-Shiny

Web Technologies: Angular, React, Spring Boot, JWT, Spring MVC, Maven, RESTful services

Development Practices: Object Oriented Programming, Microservice Architecture, Data Structures, CI/CD

Tools and Framework: Tableau, Microsoft Power BI, Snowflake, DynamoDB, MS Excel, R-Studio, Google Collab, VS Code, Android Studio, Git, GitHub, TensorFlow, Transformers, Docker, Pandas

AI Tools & LLMs: Lovable, Cursor, Windsurf, Julius AI, ChatGPT, Gemini, Grok, Claude

Certifications: Amazon Web Services Certified Cloud Practitioner | [Credentials](#)